Q&A LECTURE

Introduction to Digital Trace Data:
Quality, ethics, and analysis

Q&A lecture

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Exam and presentations

Grade of the presentation: individual grades are added to OSIRIS next Wednesday.

EXAM

- **Goals of the course:** understand and identify problems in data collection and analysis.
- Anything we covered in the lectures and the labs (and related material from readings).

19 **multiple choice questions**

Answer all questions, even if you have to answer at random.

3 **open text questions reflecting scenarios**

- Advantages and disadvantages of different approaches
- Type of errors (representation side or measurement side)
- Ethical principles

Read the email Javier sent last week, and bring your ID!

Student questions

1

Does it require close reading of the literature?

In some cases yes, but be critical with details. Are they relevant for the course?

2

What kinds of questions will be there?

You will see some examples today.

3

Will there be questions about coding?

No.

WEEK

1

Digital trace data

How do we understand human behaviour and societies → and what does DTD add?

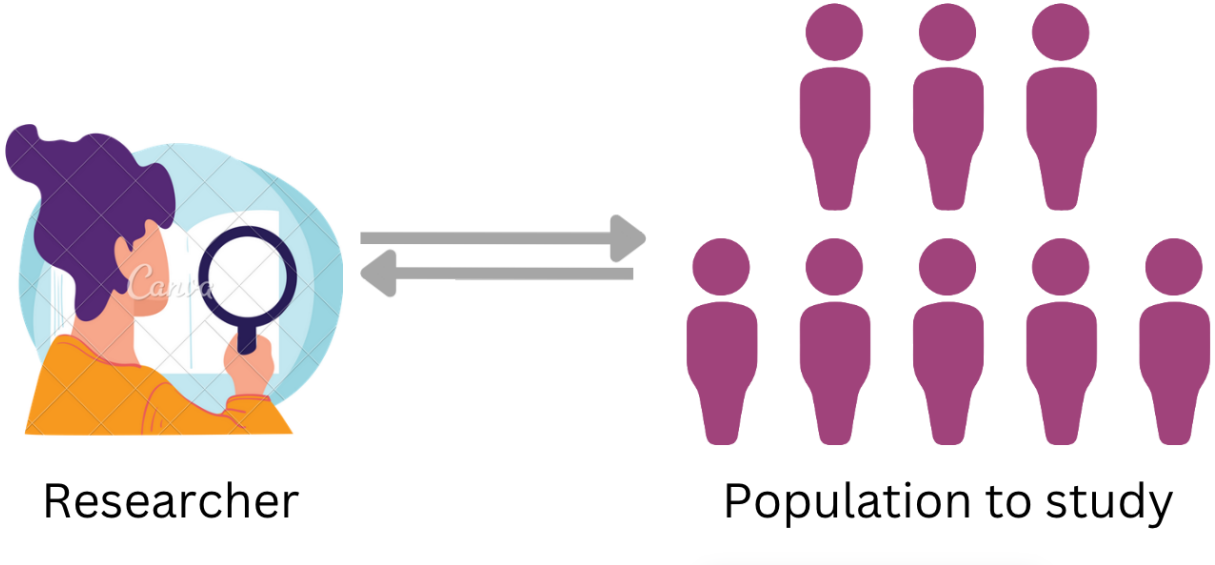
How do we understand human behavior/societies?

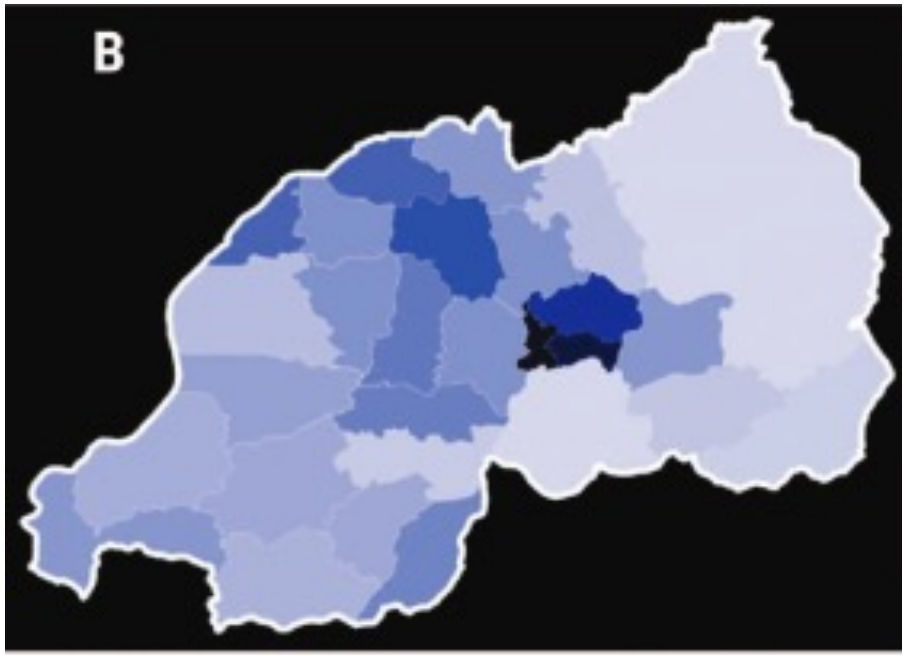
e.g. determining poverty in Rwanda



OUR TRADITIONAL APPROACH

ASK QUESTIONS



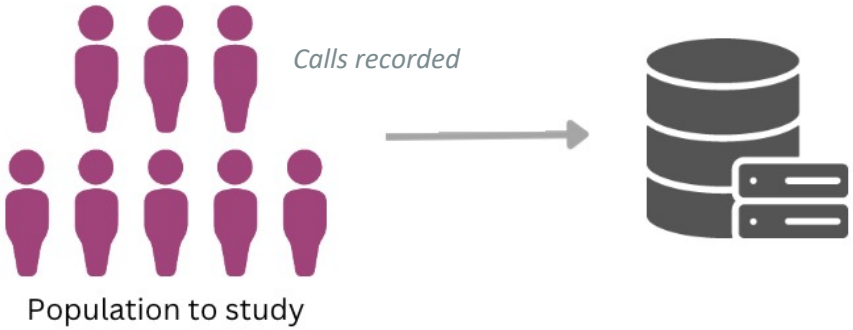


(Blumenstock et al., 2015)

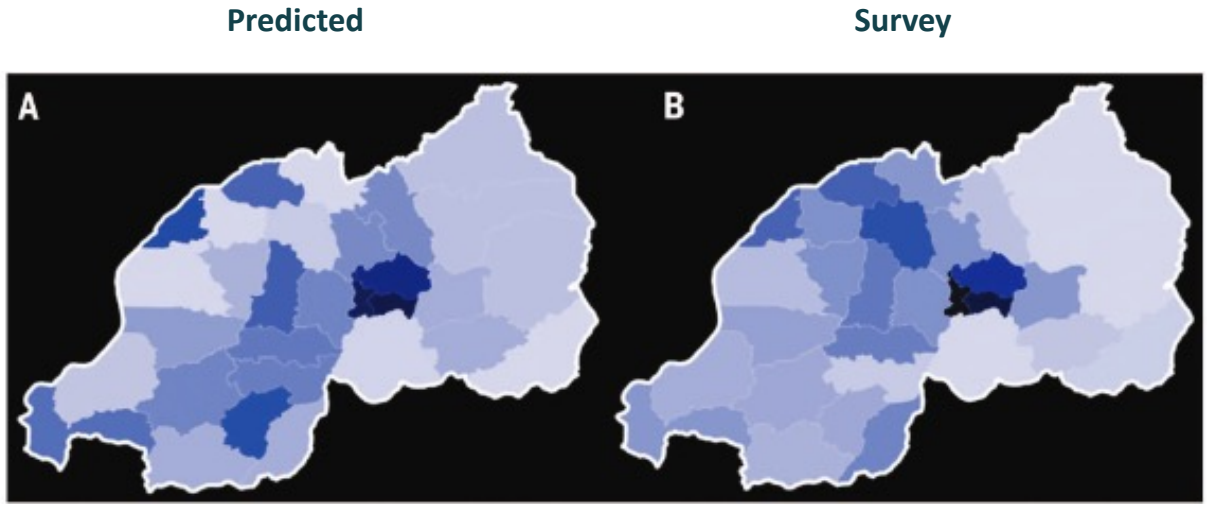
How do we understand human behavior/societies?

USING DIGITAL TRACE DATA

But we could also use the records of individuals' digital activities, such as phone call records.







(Blumenstock et al., 2015)

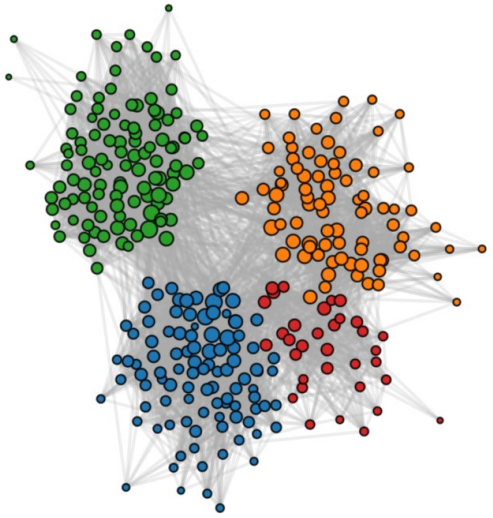
Advantages and disadvantages of DTD

Advantages

- **Unprecedented level of granularity:** study small groups
- **Always on:** longitudinal data (dynamics! historical!)
- **Non-reactive:** study people “in-the-wild” (self-reported and real behaviour differ)
- **Cheaper** than surveys
- **New research possible:** e.g. social interactions

Disadvantages

- **Prone to errors:** measurement / representation
- Confounded by algorithms
- Incomplete
- Drifting
- Dirty
- Sensitive
- Inaccessible



WEEKS

2–3

Collecting data

User-centric and platform-centric approaches

Collecting data (weeks 2 and 3)

Differences between user-centric and platform-centric approaches. Advantages/disadvantages of each approach.

User-centric design week 2

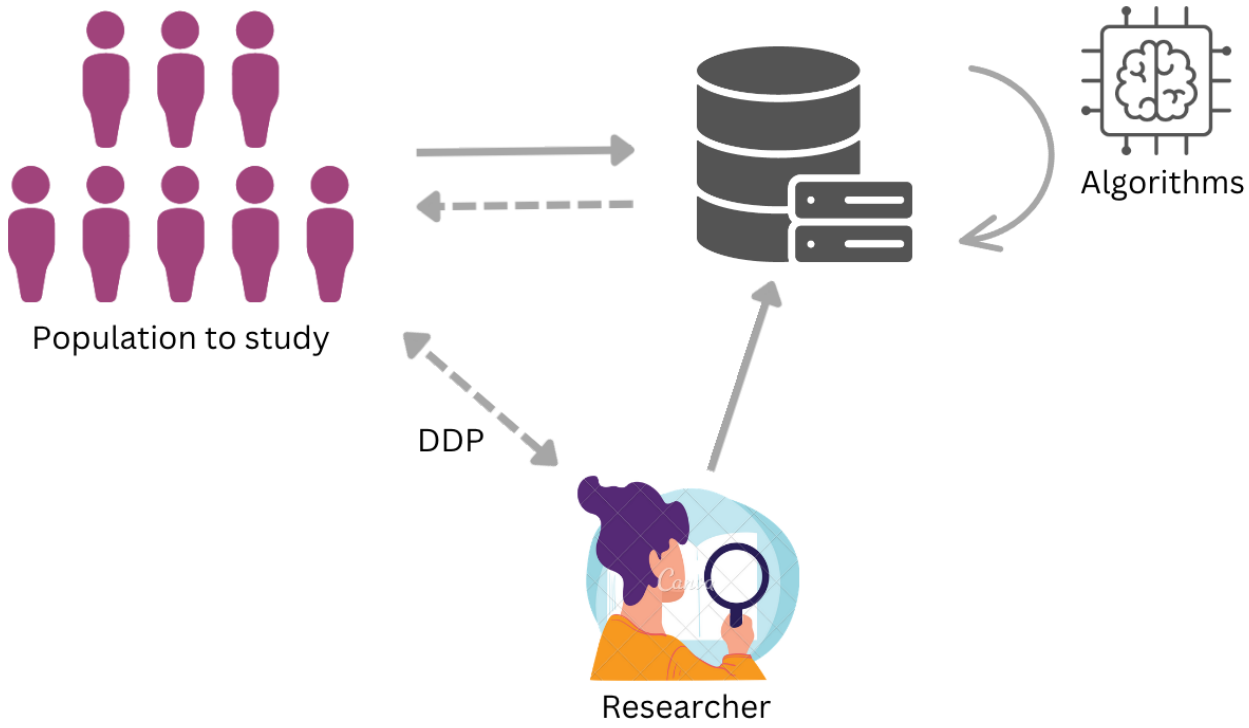
Data flows from the population to the researcher (e.g. via DDPs).

Platform-centric design week 3

The researcher collects data from the platform's database.

Algorithms week 5

Both the platform data and the population are shaped by algorithms.



Collecting data (weeks 2 and 3)

USER-CENTRIC

Sensors, apps and data donation

PLATFORM-CENTRIC

APIs, web scraping and DSA

GDPR

Enables data donation approaches

And creates restrictions:

- Collect data with a legal basis (informed consent *or* legitimate interest)
- Purpose / data / storage limitation
- Accuracy / security / accountability

EXAMPLE MC QUESTION

What is the main purpose of an API key accessing web APIs?

a. To authenticate and authorize access to the API, often tracking usage and enforcing rate limits

b. To bypass rate limits and access unlimited data from the API

c. To provide direct access to a database without authentication

d. To automatically download data without restrictions

EXAMPLE MC QUESTION

Why might a researcher choose a platform-centric design over a user-centric design?

a To collect data with minimal consideration for participant consent

b To obtain data from a large, diverse population without needing individual participation

c To ensure the measurement errors are minimized

d To obtain highly personalized data supplemented with self-reported information

WEEK

4

Errors in digital trace data

Representation, measurement and designed big data

STUDENT QUESTION

Can you give more examples on measurement and representation errors?

Example: you are interested in measuring affective polarization in Dutch society.

Representation

Is your *data* representative of the *target population*?

Twitter users using a hashtag

↓

Dutch population

Measurement

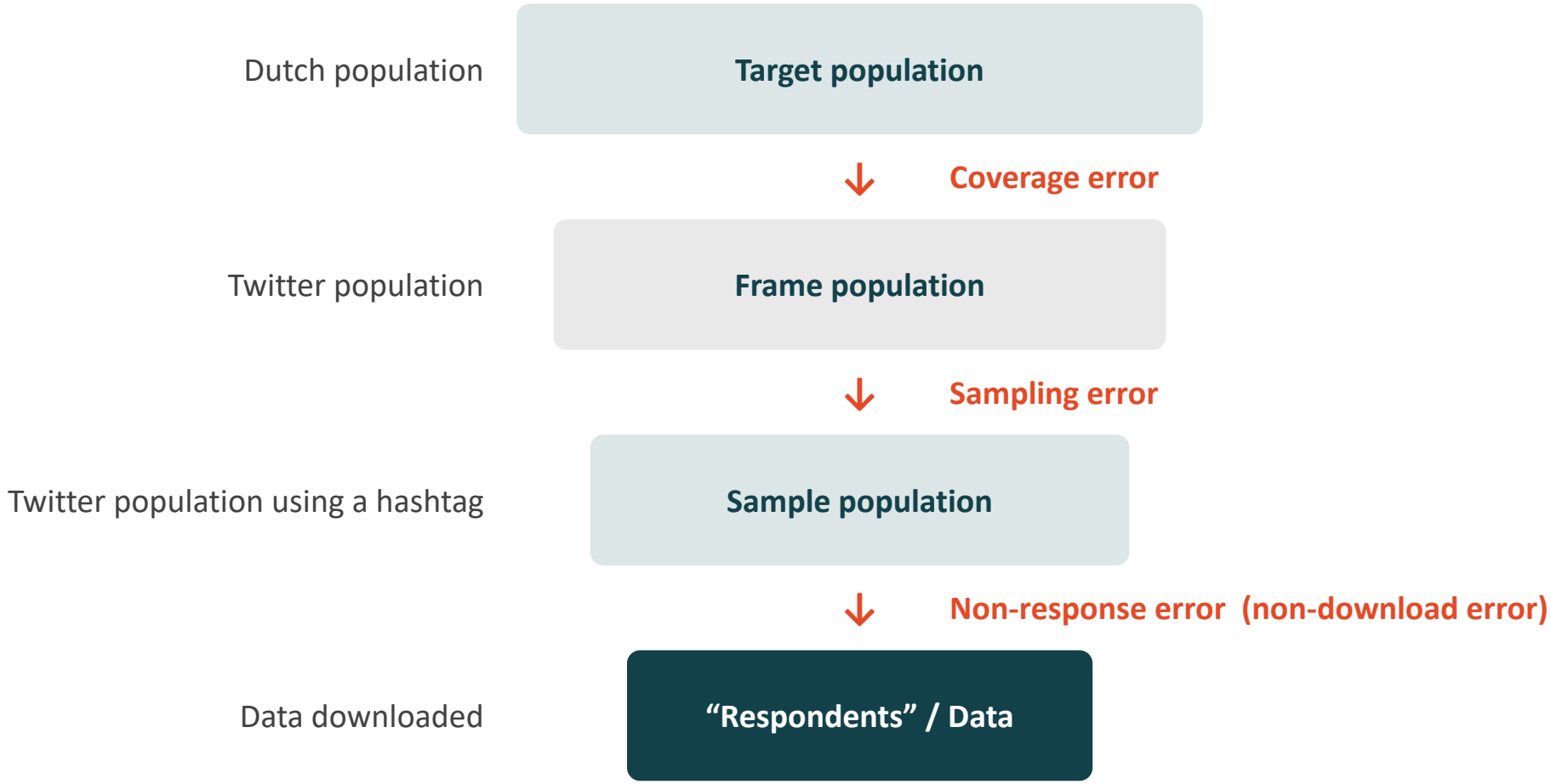
Do your *variables* measure *what you are interested in*?

Number of likes from left-wing and right-wing users

↓

Affective polarization

Errors (week 4): representation, platform-centric



Errors in DTD (week 4): measurement

MEASUREMENT

Validity

- Are likes on Twitter a good measurement of affective polarization?
- Are Facebook friends a valid measurement of real social networks?

MEASUREMENT

Measurement error

- Are the measurements correct?
- (maybe only the first 10 friends can be extracted)

MEASUREMENT

Processing error

- Extraction, integration, labeling, etc.

EXAMPLE MC QUESTION

In a study asking participants to install an app that tracks sleep patterns, which of the following is NOT considered a form of nonresponse error?

a. Some participants never install the app after being invited

b. Some participants stop using the app halfway through the study

c. Some participants skip the daily survey questions in the app

d. People without smartphones are excluded from being invited to participate

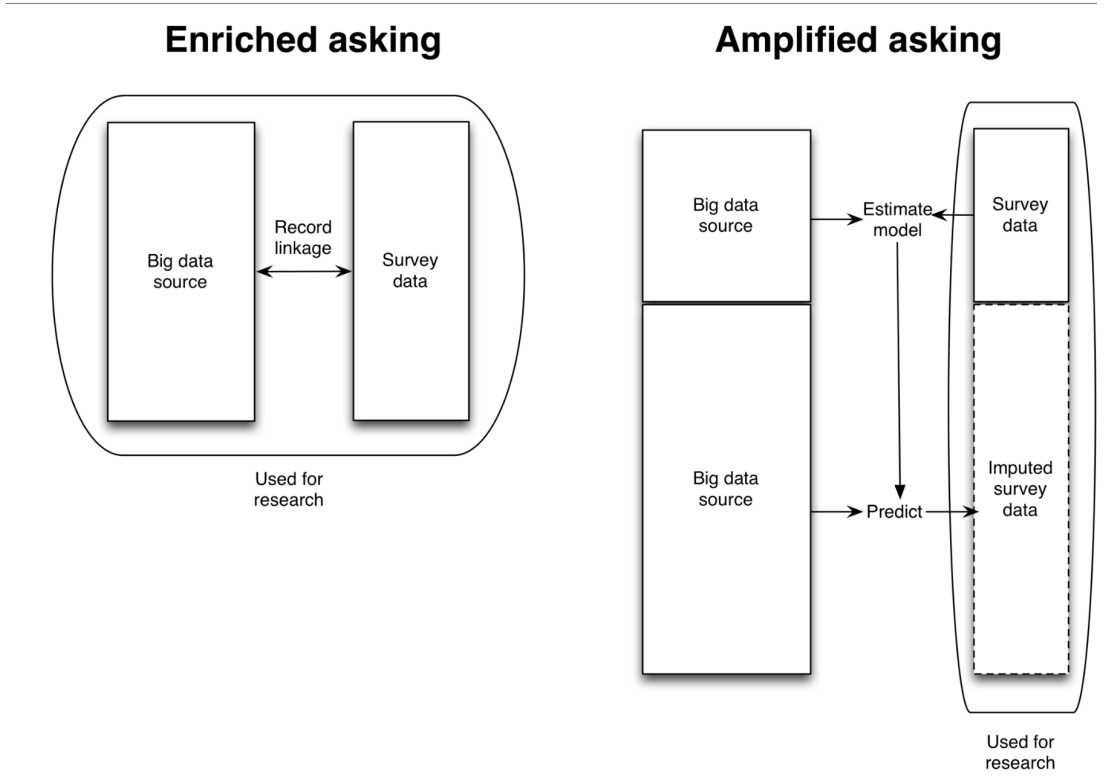
Designed big data (week 4)

MERGING SURVEY AND DTD

- Answering questions that cannot be answered with only one type of data (**enriched asking**)
- Creating predictions for a much larger population (**amplified asking**)

NOTE

It can also help correct for errors of representation.



EXAMPLE MC QUESTION

Which of the following scenarios best illustrates amplified asking?

a. A sociologist merges social survey data with census records to explore demographic patterns in social attitudes

b. A researcher combines school attendance data with student test scores to understand the relationship between attendance and academic performance

c. A healthcare researcher links individual-level data from a survey on mental health with local crime statistics to estimate the mental health impact of neighborhood safety on a larger population

d. An economist links employment and job satisfaction data to investigate the relationship between job stability and employee well-being

WEEK

5

The role of AI

Machine learning, power, fairness and bias

The role of AI (week 5)

AI IS USED

By companies
e.g. recommendation systems

→ **Validity problems** (measurement)

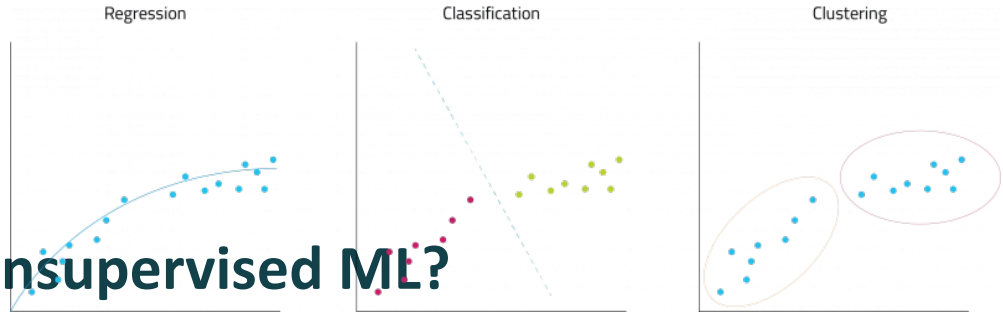
By research
in the labeling process

→ **Processing error** (measurement)

? **Are these measurement or representation errors? Why?**

STUDENT QUESTION

Can you give more examples of supervised and unsupervised ML?



Supervised

We have inputs (features, independent variables) and an output (target, dependent variable).

EXAMPLES

- **Email spam detection:** the model learns from labelled emails (“spam” or “not spam”) to make predictions for new emails coming in.
- **House price prediction:** the model learns from past data (“house size”, “location” and outcome “price”) to estimate future house prices.

Unsupervised

We have inputs and (mostly) try to find groups.

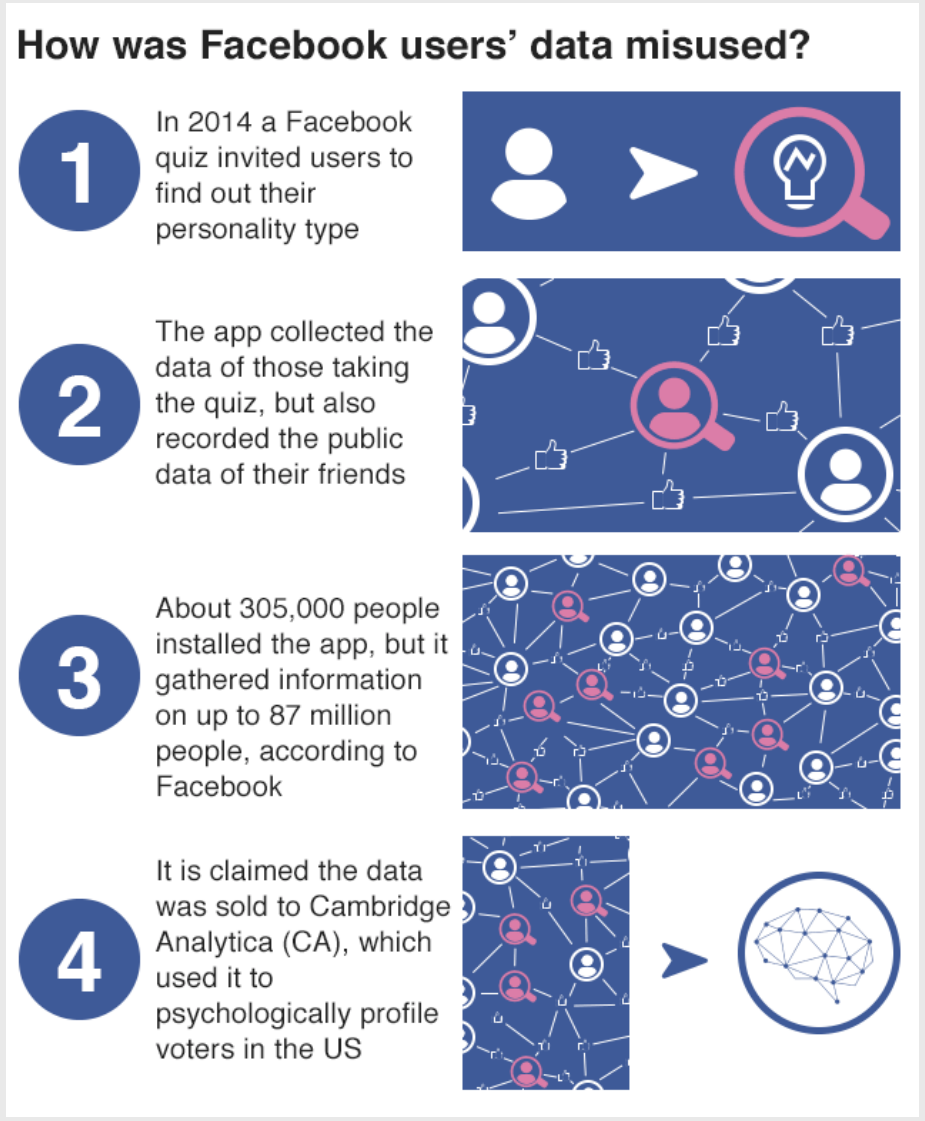
EXAMPLES

- **Customer segmentation:** grouping shoppers into clusters based on their behavior, without predefined labels.
- **Topic modelling:** discovering themes in a large set of text documents without knowing the topics beforehand.

Examine power (week 5)

EXAMINE POWER

1. Who is involved in creating AI (and who is not)?
2. Whose goals/interests are prioritized (and whose are not)?
3. And who benefits (and who is either overlooked or actively harmed)?



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STUDENT QUESTION

What is the exact difference between the structural domain and the disciplinary domain in the matrix of domination? Do you have an example?

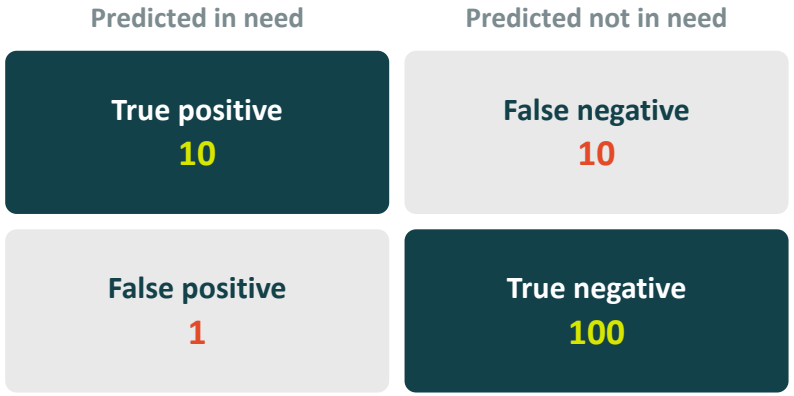
Table 1.1: The four domains of the matrix of domination[14]	
Structural domain Organizes oppression: laws and policies.	**Disciplinary domain** Administers and manages oppression. Implements and enforces laws and policies.
Hegemonic domain Circulates oppressive ideas: culture and media.	**Interpersonal domain** Individual experiences of oppression.

DOMAIN	DESCRIPTION	EXAMPLE
Structural (the setup of a system)	Overarching system and institutions that organize access to power and resources	Credit laws and systems historically exclude women and racialized groups
Disciplinary (how the system enforces rules and norms daily)	Mechanisms that enforce norms, control and surveillance within those systems	Credit scoring algorithms penalize those same groups and limit opportunities

The role of AI (week 5)

- ML models often yield errors different for different subpopulations (i.e., **lack of fairness**).
- We can measure the quality of the model (for different subpopulations) using the **confusion matrix**.

ASSISTIVE

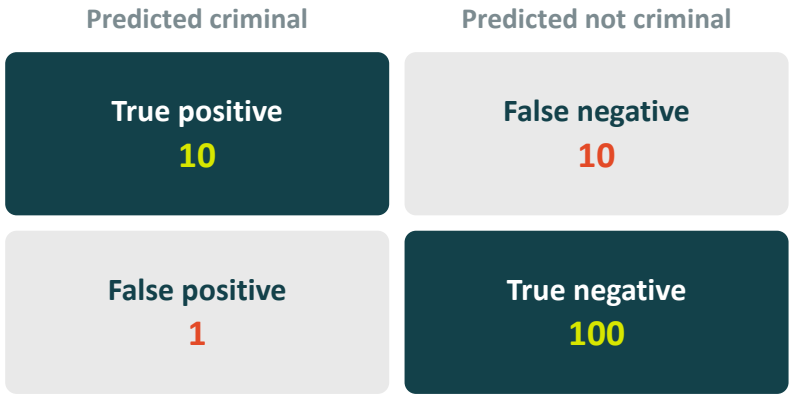


IN GENERAL

Assistive interventions: false negatives should be avoided (but also false positives if the budget is limited).

Punitive interventions: false positives should be avoided.

PUNITIVE



The role of AI (week 5): sources of bias

Main idea: if the model was trained on biased data → the model will be biased when you use it.

> *e.g.* A company uses an ML model to predict whether a job candidate will be a successful employee based on their employment history, education level, and criminal record.

SOURCE OF BIAS	EXAMPLE	IMPACT
Sample bias Certain groups are (vastly) underrepresented in the training data	The training data has limited examples from international candidates	The model may perform poorly on these groups, leading to biased hiring recommendations
Feature bias Certain features have different meaning for different subpopulations	"Educational background" may favour candidates from prestigious universities, which are less accessible to lower-income groups	The model may unfairly favour candidates from privileged backgrounds, reinforcing inequalities
Outcome bias The outcome has different meaning for different subpopulations	Success is defined by promotion history, which may be biased against historically marginalized groups due to prior discrimination in the workplace	The model perpetuates past inequalities, overlooking qualified candidates from diverse backgrounds
Pipeline bias Errors in data processing and model training	e.g. removing candidates with international work experience	
Application bias Biases in how the model is applied	e.g. trusting an AI hiring men, but not to other genders	

The role of AI (week 5): sources of bias

Main idea: if the model was trained on biased data → the model will be biased when you use it.

e.g. A researcher uses a machine learning model to label a large dataset of social media posts to understand public sentiment on a new public policy. The model is trained on text from a right-wing forum labeled by UU students.

SOURCE OF BIAS	EXAMPLE	IMPACT
Sample bias Certain groups are (vastly) underrepresented in the training data	The model is trained on a right-wing forum	May be biased for other groups
Feature bias Certain features (the words, in case of text) have different meaning for different subpopulations	"Woke" may be seen as something negative for right-wing people and positive for left-wing people	The model may classify posts containing "woke" as negative
Outcome bias The outcome has different meaning for different subpopulations	UU students may be more left-wing than average and label as negative posts that were not intended to be negative	The model reflects the views of the labellers
Pipeline bias Errors in data processing and model training	e.g. removing slang	
Application bias Biases in how the model is applied	e.g. the researcher re-labels the social media posts of particular groups	

STUDENT QUESTION

Can you explain the difference between feature and outcome bias?

A researcher uses a machine learning model to label a large dataset of social media posts to understand public sentiment on a new public policy. The model is trained on text from a right-wing forum labeled by UU students.

Feature bias

Certain features have different meaning for different subpopulations

- Things intended as negative are classified as positive, or vice versa.

Outcome bias

The outcome has different meaning for different subpopulations

- Classifications of the model reflect the views of the labellers and not of the population: bias in the outcome when trying to understand the public sentiment.

→ **Difference is small but important, two different contexts!**

STUDENT QUESTION

Can you explain the difference between feature and outcome bias … in the context of predictive policing?

Feature bias

Certain features have different meaning for different subpopulations

- If police patrol some neighbourhoods more often, there will be more recorded arrests there, even if the people don't commit more crimes.
- The algorithm learns that those areas are "high crime", which reinforces over-policing.

Outcome bias

The outcome has different meaning for different subpopulations

- If the model predicts where arrests happen instead of where crimes occur, it is using a biased outcome, because arrests depend on who gets policed, not who commits crimes.
- The model ends up predicting police activity, not real criminal activity.

EXAMPLE MC QUESTION

Which of the following best describes fairness in machine learning model predictions?

a When the model's prediction error rates are similar across different demographic groups

b When the model consistently predicts higher probabilities for marginalized groups

c When the model maximizes accuracy regardless of subgroup differences

d When the model uses fewer features to minimize computation time

WEEK

6

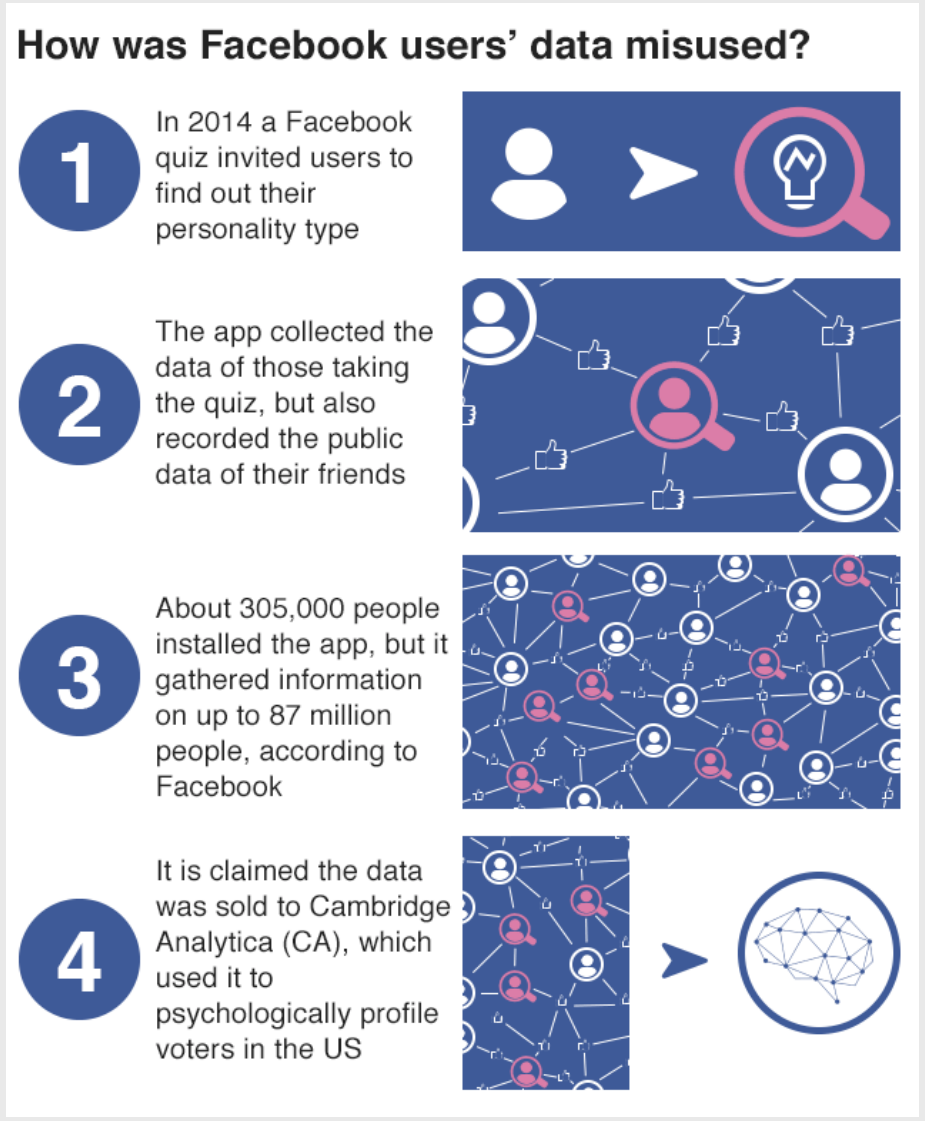
Ethics

Principles, challenging areas and ethical frameworks

Ethics (week 6)

FOUR PRINCIPLES

1. Respect for persons
2. Beneficence
3. Justice
4. Respect for law and public interest



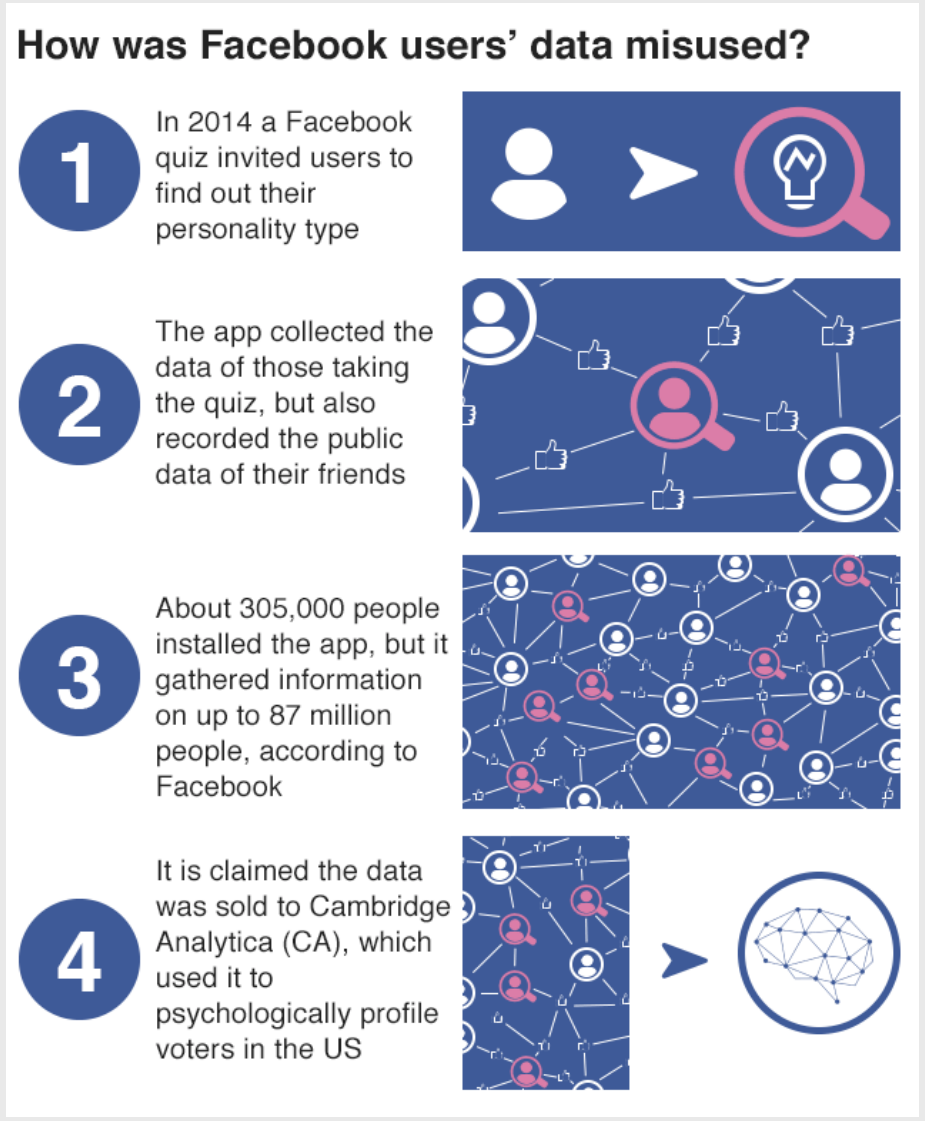
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Ethics (week 6)

FOUR CHALLENGING AREAS

- **Informed consent**
- **Managing informational risk**
- **Privacy:** appropriate flow of personal information
  - Actors, attributes and transmission principles of contextual integrity
- **Making decisions in the face of uncertainty**
  - Precautionary principle: “Better safe than sorry”

i **The GDPR principles provide guidance!**



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Ethics (week 6): ethical frameworks

A

Deontology

The ***means*** matter most

B

Consequentialism

The ***ends*** matter most

C

Virtue ethics

Be ***good / virtuous***

EXAMPLE MC QUESTION

Which ethical framework emphasizes the importance of fostering honesty and kindness in individuals rather than focusing strictly on duties or outcomes?

a Virtue ethics

b Deontology

c Consequentialism

d Utilitarianism

EXAMPLE OPEN QUESTION

CASE · ECOTALK

In January 2023, two researchers, Sarah Mills and David Cho, conducted a study on public opinions regarding climate change by gathering data from the popular social media platform EcoTalk. EcoTalk is widely used for discussions on environmental issues and makes user comments and reaction data publicly visible. Mills and Cho accessed the platform's paid API, which allowed them to retrieve data, including usernames, post timestamps, user reactions (likes, shares), and comment text. They collected over 200,000 comments from users across various regions over two years.

After completing their study, Mills and Cho decided to publicly release the dataset to "support further research in public sentiment on climate change."

QUESTIONS

a Compare briefly the advantages and disadvantages of using an API versus a data donation approach in this example. Identify 2–3 advantages of each method.

b Discuss the legal considerations Mills and Cho would need to address if they had chosen to scrape the data instead of using the API.

c List the four ethical principles and provide a reflection on how each principle applies to Mills and Cho's decision to release the EcoTalk dataset.

Outlook

1

Vast possibilities

DTD open vast possibilities for understanding societal trends, with the power to address global challenges if used ethically. But a lot can go wrong if the data is used carelessly.

2

What you know now

How to ethically collect data and properly understand the errors! You also know that data and models are not neutral but can consolidate or break down power relations.

3

Next steps

Continue being critical. Learn how to analyze data (e.g. Applied Data Science and Visualization course, Text Mining course).



Other questions?

Laura Boeschoten · Department of Methodology and Statistics

Good luck with the exam!

Please take a moment to fill in the course evaluation

caracal.uu.nl

EVALUATION